Mastering Value Categories with Standard Tools

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Who Am I?

- A C++ Developer, working at SolarEdge
- Active member of the ISO C++ work group (WG21):
  - Israeli NB Chair
  - Ranges SG Chair
  - Assistant LEWG Chair

- I love software design
- I also have a passion for cataloguing, which is the main reason I'm giving this talk (as well)

Outline

- **Part I:** Intro to value categories
- **Part II**: Value categories in practice
- **Part III**: Value categories and generic code
- **Part IV**: Manipulating value categories
- **Part V:** Advanced utilities

- **Disclaimers:**
  - Sometimes I “cut corners”, full definitions are in cppreference.com and in the standard
  - Examples will (mostly) elegantly ignore function pointers and arrays, and focus on objects

Part I: Intro To Value Categories

- What are value categories
- Evolution of value categories

Part I: Intro To Value Categories

- Let's start with usability:

```
struct Data {
   Data(size_t s);        // Constructor
   Data(const Data&);     // Copy
   Data(Data&&);          // Takes ownership

   int* buff;
   size_t s;
};

const Data getData(size_t s) {
   return Data(s);
}
```

```
auto d1 = Data(1024);                  // 1
auto d2 = std::move(d1);               // 0
auto d3 = getData(1024);               // 1
auto d4 = std::move(getData(1024));    // 2 ☹
```

1

Part I: What Are Value Categories

- Value Categories were inherited from C, with the porting of “lvalue expression”

- The original term referred to the location of the expression in regard to assignment

```
auto a = int(42);
```

  - lvalue (left-value) was on the left of the assignment
  - rvalue (right-value) was on the right of the assignment

- Value Category of an entity defines:
  1. Its **lifetime** (can it be moved from, temporary, observable after change, etc.)
  2. Its **identity** (object has identity if its address can be taken and used safely)

- Value Categories affect two very important aspects relevant to libraries:
  1. Performance
  2. Overload resolution

Part I: What Are Value Categories

- Value Category is a quality of an **expression**

```
struct Data {
    Data(int x);
    int x_;
};
void foo(Data&& x) {
    x = 42;
}
```

```
Data&& a = 42;
foo(a);              // Fail! lvalue!
foo(Data(73));       // OK
```

- What (misleadingly) looks like the value category, **can in fact be the type**
  - **a's Type:** rvalue reference to Data
  - **a's Value Category:** lvalue

- What (misleadingly) looks like the same entity, **can in fact have a different VC in different contexts**
- During a function call:
  - **Step I:** Calls constructor, creates an unnamed temp `Data(73)`
  - **Step II:** `Data(73)` binds to the rvalue reference x
  - **Step III:** The entity which *used to be* `Data(73)` has a name, therefore it is now lvalue

Part I: What Are Value Categories

- Each **expression** has two properties:
  1. A type (including CV qualifiers)
  2. A value category

- Value Category is a quality of an **expression**

(*)

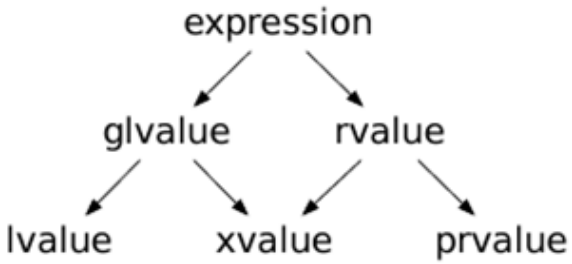


Figure 1 — Expression category taxonomy

(*) Introduced in: N3055: A Taxonomy of Expression Value Categories (William M. Miller, 2010)

- VCs changed dramatically throughout the lifetime of C++ versions, affected by the rules defining **references**, **move semantics** and **copy elision**.

Part I: Evolution Of Value Categories

- C language:
  - Three types of expressions:
    - lvalue expression
    - Non-lvalue object expression
    - Function designator expression

- C++98 [ISO/IEV 14882:1998]: added (lvalue) references:
  - Expression is either an lvalue or an rvalue:
    - Lvalue: Objects, Functions, References
    - Rvalues: Non-lvalue (can be bound by const lvalue reference)

- C++03 [ISO/IEC 14882:2003]
  - No significant change

Part I: Evolution Of Value Categories

- C++11 [N3242]: added rvalue references, move semantics:
  - By: Howard E. Hinnant, Peter Dimov, Dave Abrahams
  - N1377: A Proposal to Add Move Semantics Support to the C++ Language (2002)
  - (N1385: The Forwarding Problem: Arguments (2002))
  - N2118: A Proposal to Add an Rvalue Reference to the C++ Language (2006)

	Has Identity (glvalue)	Doesn't have identity
Can't be moved from	lvalue	-
Can be moved from (rvalue)	xvalue	prvalue

- C++17 [N4659]: Added guaranteed copy elision:
  - P0135: Guaranteed copy elision through simplified value categories (Richard Smith)

	Has Identity (glvalue)	Doesn't have identity
Can't be moved from	lvalue	-
Can be moved from (rvalue)	xvalue	prvalue's materialization

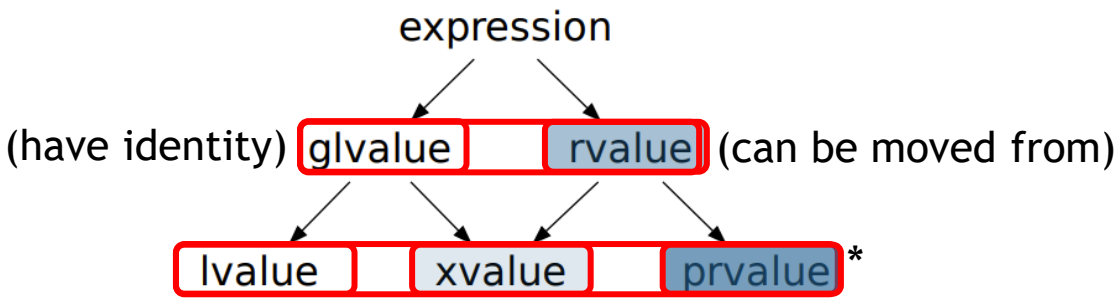
"The ***result of a prvalue*** is the value that the expression stores into its context"

Part I: Evolution Of Value Categories

- C++20:
  - [N4861] (March 2020):
    - P0527: Implicitly move from rvalue references in return statements (David Stone)
    - Editorial: Moved Value Categories section from [basic] to [**expt**]
  - [N4868] (Oct 2020):
    - Removed “bit-field” from the value categories primary definitions

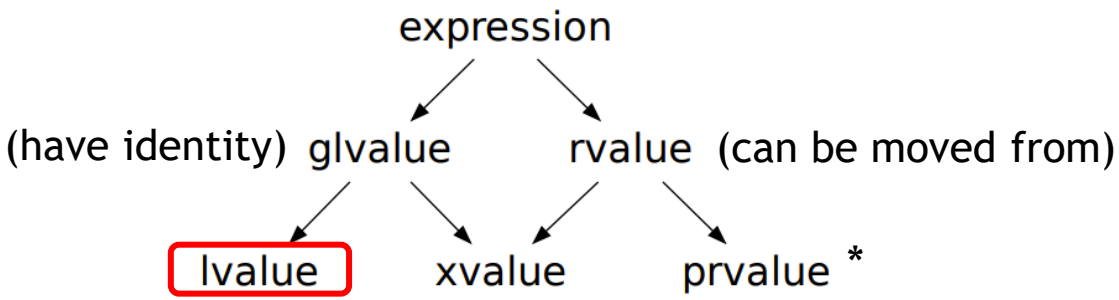
- C++23 draft (latest):
  - P0847: Deducing this (Gašper Ažman, Sy Brand, Ben Deane, Barry Revzin)
  - (P0847 also added like_t)
  - P2445: std::forward_like

Part I: Evolution Of Value Categories



- Main categories (classification only)
  - **glvalue**: expression *whose evaluation determines the identity* of an object or function
  - **rvalue**: a prvalue or an xvalue
- Subcategories
  - **lvalue:** glvalue that is not an xvalue
  - **xvalue:** glvalue that denotes an object whose resources can be reused (usually because it is near the end of its lifetime)
  - **prvalue:** expression *whose evaluation initializes* an object, or computes the value of the operand of an operator, as specified by the context in which it appears, or an expression that has type cv void

Part I: Evolution Of Value Categories



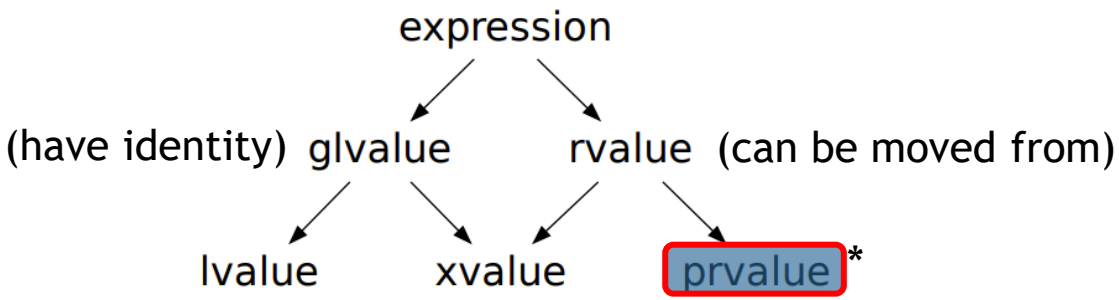
- Examples:

```
struct Data { int n; int* pn = &n; };
Data& getData(Data& d) { return d }
int a = 42;
int b = a;
int& la = a;    // ra = a = b;
int* pa = &a;
int&& ra = 42;
a++;
++a;
```

```
int arr[] = {1, 2, 3};
arr[0] = 73;
Data d;
(&d)->n = 42;
d.n = 73;
*d.pn = 42;
string s = "Hello World";
a==b ? b : c; // when b, c are lvalues
Data c = getData(d);
```

(*) ra has the **type: rvalue reference to int**, with the **value category: lvalue**

Part I: Evolution Of Value Categories

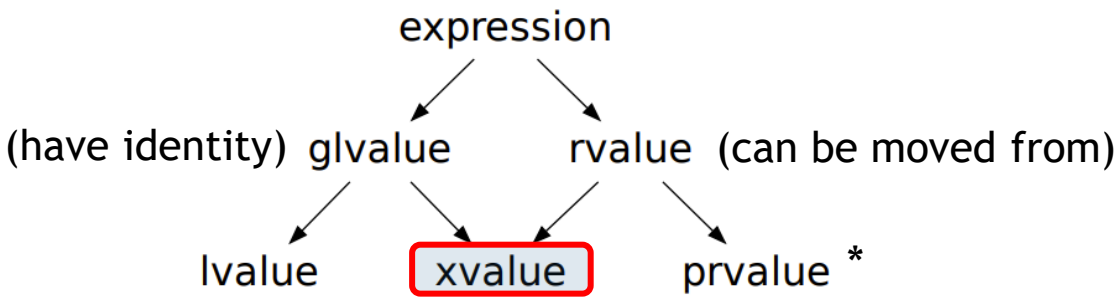


- Examples:

```
struct Data {
    int n;
    int foo() { this->n = 4; }
};
int a = 42;
int* pa = &a;
pa = nullptr;
a++;
++a;
```

```
auto l = []() { return 2; };
Data d;
Data* dp = &d;
d->n = 6;
d.n = 6;
string s = "Hello World";
a==a ? throw 4 : throw 2;    // void
bool equals = a==42;
```

Part I: Evolution Of Value Categories



- Examples:

```
struct Data { int n; int* pn = &n; };


Data d1 = Data(42);
d1.*n = 73;
Data d2 = std::move(d1);
```

```
Data getData() {
    return Data(73);
}

Data d3 = getData();
d1==d2 ? Data(42) : Data(73);
```

Part II: Value Categories In Practice

- Intro to overload resolution
- The details of binding
- Value categories in generic code

Part II: Intro To Overload Resolution

- Overload resolution of functions affected by value categories
- To compile a function call, compiler creates list of candidates, then finds correct overload

- **Overload Resolution:**
  - Picking the correct function from group of candidates
  - Function candidates are picked according to function name (Name Lookup)
  - May include Template Argument Deduction (TAD) and Argument Dependent Lookup (ADL)

- **Template Arguments Deduction (TAD):**
  - Value category of an expression affects type deduction
  - Involves non-trivial deduction rules, which are **out of scope** for this talk (e.g. deduction of T from multiple params with different value categories)

Part II: Intro To Overload Resolution

- **To summarize:**
  - To compile a function call, compiler must first perform name lookup, which may involve argument dependent lookup, and for function templates may be followed by template argument deduction.
  - If these steps produce more than one *candidate function*, then overload resolution is performed to select the function that will be called.
  - If result is ambiguous (= multiple candidates with same level of fit), compilation fails.

Part II: The Details Of Binding

- Expressions with different Value Categories "bind" to different **types of References**

- The Reference type **which binds the expression** determines the permitted operations

```
int a = 42;
int& la = a;
const int& cla = a;
int&& ra = a + 73;
const int&& cra = 42;
```

```
la = 73;      // OK, a = la = 73
cla = 42;     // Error!
ra = 73;      // OK
cra = 42;     // Error!
```

- Binding rules are important as part of the following "events":
  1. Initialization or assignment
  2. Function call (including non-static class member function called on an object)
  3. Return statement

Part II: The Details Of Binding

1. Initialization Or Assignment:
   As part of reference initialization, the following apply:
   - **lvalue reference**: binds lvalues only
   - **const lvalue reference**: binds both **lvalues** and **rvalues**
   - **rvalue reference:** binds rvalues only
   - **const rvalue reference**: binds rvalues only

```
int a = 42;
int& la1 = a;                    // OK, la = 42
int& la2 = 73;                   // Error
const int& cla1 = a;             // OK
const int& cla2 = 73;            // OK
int&& ra1 = a;                   // Error
int&& ra2 = a + 42;              // OK
const int&& cra1 = a;            // Error
const int&& cra2 = a + 42;       // OK
```

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- The Lifetime of an object can be extended by binding to references:
  - **const lvalue reference**: extends lifetime of an object, not allowing modification
  - **rvalue reference**: extends lifetime of temporary objects

Part II: The Details Of Binding

2. Function Call:

```
struct Data {
   Data(int n) : _n(n) {}
   int _n;
};
const Data getData(int x) {
   return Data(x);
}
void foo(Data& x) {}          // 1
void foo(const Data& x) {}    // 2
void foo(Data &&x) {}         // 3
void foo(const Data &&x) {}   // 4
```

```
Data &ld = d;
const Data& cld = d;
Data&& rd = Data(73);
const Data&& crd = Data(42);
foo(ld);                      // lvalue: 1,2
foo(cld);                     // const lvalue: 2
foo(rd);                      // lvalue: 1,2
foo(crd);                     // const lvalue: 2
foo(Data(73));                // xvalue: 3,4,2
foo(getData(42));             // const xvalue: 4,2(!)
```

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(*) Example scheme by Boris Kolpackov

- Limitations on the object in the context of the function are according to the **binding function**:
  - **lvalue reference**: Function can modify data, Caller can observe data changes
  - **const lvalue reference**: Function can't modify data, Caller can observe (old) data
  - **rvalue reference**: Function can modify the data, Caller shouldn't observe data
  - **const rvalue reference**: Function can't modify data, Caller shouldn't observe (old) data (*)

- Non-static member functions behave similarly 2
  (*) deleted const rvalue reference overload will **block** rvalue binding

Part II: The Details Of Binding

3. Return Statement:
   - Starting from C++17, the behavior of VCs is affected by: "P0135: Guaranteed copy elision (…)"
   - There are two mandatory elisions of copy and move constructors:
     1. Object initialization

```
Data d = Data(Data(42));                    // 1 CTOR (avoids: CTOR, Copy CTOR)
```

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     2. Return statement:
        An un-named Return Value Optimization (RVO):

```
Data getData(int x) {
    return Data(x);
}
```

```
Data d = getData(42);   // 1 CTOR (avoids: CTOR, Move CTOR)
```

(*) No change in *non-mandatory* Named Return Value Optimization (NRVO):

```
Data getData(int x) {
    Data d(x);
    return d;
}
```

```
Data d = getData(42);   // 1 CTOR (avoids: CTOR, Move CTOR)
```

Part II: The Details Of Binding

3. Return Statement: **Materialization**

Temporary materialization conversion [conv.rval]

A prvalue of type T can be *converted* to an xvalue of type T.
This conversion initializes a temporary object of type T from the prvalue by ***evaluating the prvalue*** with the temporary object as its result object, and produces an xvalue denoting the temporary object.
[in order to materialize] T shall be a complete type.

```
Data getData(int x) {
    return Data(x);
}

Data a = getData(42);
```

1. prvalue of type Data
2. Temp Data evaluating prvalue
3. Produces an xvalue

```
Data getData(int x) {
    Data d(x);
    return d;
}

Data a = getData(42);
```

1. lvalue of type Data
2. lvalue initializes temp Data
3. Produces an xvalue

Part II: The Details Of Binding

3. Return Statement: **Const vs. Non Const**

```
Data get(int x) {
    return Data(x);
}
```

```
Data a = get(42);            // lval (No temporary, RVO)
Data& b = get(73);           // Err: xval → non-const lval ref
const Data& c = get(42);     // xval → const lval ref (*)
Data&& d = get(42);          // xval → rval ref (*)
```

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(*) extends lifetime

```
const Data get(int x){
    return Data(x);
}
```

```
Data a = get(42);            // lval (No temporary, RVO)
Data& b = get(73);           // Err: const xval → non-const lval ref
const Data& c = get(42);     // const xval → const lval ref (*)
Data&& d = get(42);          // Error: const xval → rval ref
const Data&& e = get(42);    // OK (*)
```

2

(*) extends lifetime

Part II: The Details Of Binding

- To summarize:
  Binding rules apply in the following “events”:
  1. Initialization or assignment
  2. Function call (including non-static class member function called on an object)
  3. Return statement

- Behavior of the entity:
  1. Initialization: limits are according to the reference which binds it
  2. Function call: limits inside the function are according to the overload which binds it
  3. Return statement: limits as in initialization, with additional rules due to optimizations and const

Part III: Value Categories And Generic Code

Part III: Value Categories And Generic Code

- Templates instantiation and overload resolution are affected by the VC of the expression
- When a function template participates in overload resolution, Template Argument Substitution occurs.
- Template arguments can be affected by lvalue-to-rvalue, array-to-pointer, or function-to-pointer implicit conversions (as in auto)

Part III: Value Categories And Generic Code

- Reference Collapsing:
  In case of concatenation of multiple '&' symbols:
  - In generic code
  - In code using type aliases

- Compiler performs Reference Collapsing:

```
typedef int&  lr;
typedef int&& rr;

Template <typename T>
void foo(T t) {
   // ...
}
```

```
 int a;
 int&& b = a;          // int&
 int&&& c = a;         // int&
int&&& d = a;          // int&
int&&&& e = 73;        // int&&

 template <>
 void foo(int& t) {}  // Foo(int&)

 template <>
 void foo(int&& t) {} // Foo(int&)
```

Part III: Value Categories And Generic Code

- Forwarding parameters inside a function template takes Value Categories under consideration
- The term for them was first suggested by Scott Myers, “universal reference”
- Later referred to as “**forwarding reference**” ([temp.deduct.call/3])

- Due to TAD, “rvalue reference” has a special meaning in context of function template:

```
Template <typename T>
void foo(T&& t) {
   // Type of T here
}
```

```
int a = 42;
const int& cla = a;
int&& b = 73;
foo(a);              // int      &&  T=int&
foo(cla);            // const int&&  T=const int&
foo(std::move(a));   // int      &&  T=int
auto&& c = b;        // int      &&  T=int&
```

- T&& keeps the value category of the type the instantiation is based on

Part IV: Manipulating Value Categories

- std::is_same
- decltype specifier
- std::declval
- std::remove_reference
- std::move
- std::forward
- std::decay

Part IV: Manipulating Value Categories

- When template function is considered for overload resolution, Template Argument Substitution occurs

- The standard library and language provide different methods for querying and adjusting VCs
- The following (straight forward) type_traits are an example of such:
    1. std::is_lvalue_reference<T>
    2. std::is_rvalue_reference<T>

- The following section will use type traits, decltype, and overload resolution to determine VC

Part IV: (1): std::is_same

```
std::is_same<T, U>::value
```

- A type trait, result is accessible through `_v`
- Can be used to evaluate equality, including the value category of the expression:

```
std::is_same_v<int, int>;                    // True (1)
std::is_same_v<int&, int>;                   // False(0)
std::is_same_v<decltype(a), int>;            // True (1)
```

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- In (some) of the following examples we will use it to examine the value category of expressions. e.g.:

```
template <typename T, typename U>
struct remove_reference_is_same :
    std::is_same<typename std::remove_reference_t<T>, U>
{};
```

Part IV: (2): decltype specifier

```
decltype( expression )
```

- Evaluates *expression*, yields its **type** + **value category** (AKA the declared type)

- decltype (unlike auto) preserves value category. For an *expression* of type T:
  - If *expression* is xvalue, yields T&&
  - If *expression* is lvalue, yields T&
  - If *expression* is prvalue, yields T

- Can be used instead of a type, as a placeholder which preserves value categories

```
int&& f();
auto a = f();                // f():xvalue -> a:int
decltype(auto) b = f();      // f():xvalue -> b:int&&
```

Part IV: (2): decltype specifier

```
decltype( expression )
```

- The fine print:
  - The T prvalue doesn't materialize, so T can be an incomplete type (C++17)
  - If evaluation fails (entity is not found or overload resolution fails), program is ill-formed

- ((*expression*)) has a special meaning, and yields an lvalue expression.

```
int&& a = 42;
decltype(a) b = 42;     // type: rvalue ref to int  |  vc: lvalue
decltype((a)) c = 73;  // Error! Binding non const lvalue ref to prvalue
```

Part IV: (2): decltype specifier

```
decltype( expression )
```

- To summarize decltype() main use cases:

1. When the type is unknown (syntax is available from C++14):

```
template <typename T, typename U>
decltype(auto) Add(T t, U u) {
    return t + u;
};
```

```
template <typename T>
decltype(auto) Wrapper(T&& t) {
    // do something...
    return std::forward<T>(t);
};
```

2. To preserve the value category of the expression:

```
int&& a = 42;            // type: rvalue ref to int  |  vc: lvalue
decltype(a) b = a;       // Error! (binding rvalue ref to an lvalue ref a)
decltype(a) c = 73;      // type: rvalue ref to int  |  vc: lvalue
decltype((a)) d = a;     // type: lvalue ref to int  |  vc: lvalue
```

Part IV: (3): std::declval

```
std::declval<T>( )
```

- Utility function, produces:
  - xvalue expression T&&
  - If T is void, returns T
- Can be used with *expression* to return the expression's reference type
- Can return a non constructible or incomplete type

```
struct Type {
    int a;
    int Foo() { return 42; }
private:
    Type(){}
};
```

```
Type t;                                    // Fails
typeid(std::declval<Type>()).name();   // Type
```

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- Combined with decltype, can get the type of a member (even when Type is non constructible)

```
decltype(std::declval<Type>().a) b = 73;
```

- Shouldn't be used in an evaluated context (evaluating std::decltype is an error)

Part IV: (3): std::declval

```
std::declval<T>( )
```

- std::declval allows us to access T members, in a way preserves value categories

```
struct Type {
    int a;
    int &ra = a;
    int getA() { return int(73); };
    int& getRefA() { return ra; };
private:
    Type(int i) : a(int(i)) {}
};
```

```
std::declval<Type>().a;          // xvalue
std::declval<Type>().ra;         // lvalue
std::declval<Type>().getA();     // prvalue
std::declval<Type>().getRefA();  // lvalue
```

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- decltype and declval are often used to transform between type and instance, for example:

```
template<typename Derived>
using begin_adaptor_t = detail::decay_t<decltype(
    range_access::begin_adaptor(std::declval<Derived &>()))>;
```

Part IV: (4): std::remove_reference

```
std::remove_reference<T>::type
```

- A type trait, result is accessible through `_t`
- Produces an lvalue expression of type T

- Example: define the "remove_ref_is_same" trait

```
template <typename T, typename U>
struct remove_ref_is_same :
   std::is_same<typename std::remove_reference<T>::type, U>
{};
```

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- Evaluate the expressions:

```
int a = 42;
int& la = a;
const int& cla = a;
```

```
remove_ref_is_same<decltype(la), int>          // True
remove_ref_is_same<decltype(move(a)), int>     // True
remove_ref_is_same<decltype(throw 4), void>    // True
remove_ref_is_same<decltype(cla), const int>   // True
```

Part IV: (5): std::move

```
std::move( expression );
```

- Utility function, produces an xvalue expression T&&
- Equivalent to static_cast to an rvalue reference of type T

```
static_cast<typename std::remove_reference<T>::type&&>(t)
```

- Notice that std::move may not always do what you hoped:

```
void foo(int& x) {
    cout << "int&";
}
void foo(const int& x) {
    cout << "const int&";
}
void foo(int &&x) {
    cout << "int&&";
}
```

```
int a = 73;
int &b = a;
const int& c = a;
const int&& d = 42;
foo(std::move(b));   // int&&       -> int&&
foo(std::move(c));   // const int&  -> const int&
foo(std::move(d));   // const int&& -> const int&
```

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Part IV: (6): std::forward

```
std::forward<T>( expression );
```

- N1385: The forwarding problem: Arguments (Peter Dimov, Howard E. Hinnant, Dave Abrahams) (2002)
  Presented two issues: forwarding params, and returning result
- Utility function, preserves value category of the object passed to the template
- Uses std::remove_reference<T> to get the value type
- Commonly used combined with forwarding reference

```
void Foo(int& x) {
    cout << "int&";
}
void Foo(const int& x) {
    cout << "const int&";
}
void Foo(int &&x) {
    cout << "int&&";
}
```

```
template<class T>
void Wrapper(T&& t) {
   Foo(std::forward<T>(t));
}

template<class T>
void NFWrapper(T&& t) {
   Foo(t);
}
```

```
int a = 73;
const int& lca = a;
Wrapper(a);        // int&
NFWrapper(a);      // int&
Wrapper(lca);      // const int&
NFWrapper(lca);    // const int&
Wrapper(6);        // int&&
NFWrapper(6);      // int&
```

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Part IV: (7): std::decay

```
std::decay<T>::type
```

- Type trait, result is accessible through `_t`
- Performs the following conversions:
  1. Array to pointer
  2. Function to function pointer
  3. lvalue to rvalue (removes cv qualifiers, references) (issue for move-only types)

```
template <typename T, typename U>
struct decay_is_same :
    std::is_same<typename std::decay<T>::type, U>
{};
```

1

```
decay_is_same<int[42], int*>::value;  // True(1)
decay_is_same<int&, int>::value;      // True(1)
```

- This behavior should be familiar to you, as it resembles “auto”s behavior (“auto” performs auto-decay)

```
int arr[42];                  // int[42]
auto arr2 = arr;              // int* (not int[42])
```

Part V: Advanced Utilities

- std::reference_wrapper
- std::ref
- std::unwrap_reference (C++20)
- auto cast (C++23)
- Concepts (C++20)
- Deducing this (C++23)

Part V: (8): std::reference_warpper

```
std::reference_wrapper<T>
```

- Functional helper, produces lvalue expression which can be used in same places as a reference
- reference_wrapper provides extra safety of avoiding dangling reference
- Can even be used to create a container of objects of non-constructible or incomplete types [1] [2]

```
struct Type {
   friend Type getType(int i);
   friend ostream &operator<<(ostream &os, const Type& t) {
      cout << t.a;
      return os;
   }
private:
   Type(int a) : a(a) {}
   int a;
};
```

```
Type getType(int i) {
    return Type(i);
}
```

```
std::array arr = { getType(1), getType(2), getType(3) };
std::vector<std::reference_wrapper<Type>> v(arr.begin(), arr.end());
std::cout << "v: " << v.at(0) << '\n';                                  // v: 1
```

Part V: (9): std::ref

```
std::ref( expression )
std::cref( expression )
```

- Functional helper, takes an lvalue reference to object (overload (const) rvalue reference is deleted)
- Produces an object of type std::reference_type
- Starting from C++20, T may be incomplete type

```
std::array arr = { getType(1), getType(2), getType(3) };
std::vector<std::reference_wrapper<Type>> v;
v.push_back(std::ref(arr.at(0)));                    // (Not required)
std::cout << "v: " << v.at(0) << '\n';               // v: 1
```

```
template <typename T>
void RequireRef(T t) {
   // ...
}
```

```
int a = 5;
RequireRef(a);                    // By value
RequireRef(std::ref(a));          // By reference
```

Part V: (10): std::unwrap_reference

```
std::unwrap_reference<T>( expression )
```

- Defined both in <type_traits> and <functional>, result is accessible through `_t`
- Produces:
  - If T is a reference_wrapper<U>, type is U&
  - Otherwise, type is T

```
std::unwrap_ref_decay<T>( expression )
```

- Produces:
  - If T is a reference_wrapper<U> (possibly CV qualified), type is U&
  - Otherwise, type is std::decay_t<T>

Part V: (11): auto cast

```
auto x( expression )
```

- Proposed in: ”P0849: auto(x): decay-copy in the language (Zhihao Yuan)” (voted into C++23)
- Result is a prvalue of an object type (discards CV qualifiers)
- Can also be used combined with “decltype”

```
int a = 42;          // lvalue (int)
auto& b = a;         // lvalue (int&)
auto c(b);           // lvalue (int&)
auto(c);             // prvalue
```

1

Part V: (12): Concepts (C++20)

```
template <Concept T>
auto Func (T t);
```

- Concepts can be used to constraint the Value Category of the entity acceptable in an algorithm
- <concepts> library provides:

```
std::move_constructible
```

- Concept is satisfied if:
  - T is a reference type
  - T is an object type which can be constructed from a T rvalue

```
struct Movable { };
struct NonMovable : boost::noncopyable { };

template <std::move_constructible T>
void MCFunc(T t) {
    std::cout << "T move constructible\n";
}
```

```
Movable t1;
NonMovable t2;
MCFunc(t1);        // OK
MCFunc(t2);        // Fail
```

1

Part V: (12): Concepts (C++20)

```
template <Concept T>
auto Func (T t);
```

- <concepts> library also provides:

```
std::movable
```

- Concept is satisfied if:
  - T is an object, which also satisfies:
  - move_constructible<T>
  - assignable_from<T&, T> (using std::is_lvalue_reference, std::forward, etc.)
  - std::swappable<T>

- These, as well as all the <type_traits> are available for constructing our own concepts

Part V: (13): Deducting This (C++23)

```
template <typename T>
void Foo(this T&& t) {}
```

- P0847: Deducing this (Gasper Azman, Sy Brand, Ben Deane, Barry Revzin) - voted into C++23
- Allows specifying from within a member function the value category of the expression it's invoked on

```
struct Type {
    auto Foo() const &;
    auto Foo() &;
    auto Foo() &&;
};
```

```
struct Type {
    auto Foo(this const Type&);
    auto Foo(this Type&);
    auto Foo(this Type&&);
};
```

- Combined with the forwarding reference, we can now write all these in a single template function

```
struct Type {
    template <typename Self>
    auto Foo(this Self&& self);
};
```

Part V: (13): Deducting This (C++23)

```
template <typename T>
void Foo(this T&& t) {}
```

- "Deducing this" feature introduced two new utilities: `like_t` and `forward_like<T>(u)`

```
like_t<T,U>
```

- Applies CV and ref-qualifiers of T onto U (introduced in P0847)

```
like_t <double&, int>        // int&
like_t <const double&, int>  // const int&
like_t <double&&, int>       // int&&
like_t <const double&&, int> // const int&&
```

```
forward_like <T>(u) -> forward <like_t<T, decltype(u)>>(u)
```

- Forwards instance of type U with CV and ref-qualifiers of T (introduced in P2445)

```
int a = 5;
forward_like<double&>(a)        // int&
forward_like<const double&>(a)  // const int&
```

Summary

- **Part I**: What are value categories
  - What are value categories
  - Evolution of value categories
- **Part II**: Value categories in practice
  - Intro to overload resolution
  - The details of binding
- **Part III**: Value categories and generic code
- **Part IV**: Manipulating value categories
  - std::is_same
  - decltype specifier
  - std::declval
  - std::remove_reference
  - std::move
  - std::forward
  - std::decay
- **Part V:** Advanced utilities
  - std::reference_wrapper
  - std::ref
  - std::unwrap_reference (C++20)
  - auto cast (C++23)
  - Concepts (C++20)
  - Deducing This (C++23)

Thanks!

Thank you for listening ☺

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Would love to get your input!

References (☺)

- Proposals:
  - P0849: auto(x): decay-copy in the language (Zhihao Yuan) (C++23)
  - P2255: A type trait to detect reference binding to temporary (Tim Song) (C++23)
  - P2446: views::as_rvalue (C++23?)
  - P0792: function_ref: a non-owning reference to a Callable (Romeo, Yuan, Waterloo) (C++23?)
  - P2266: Simpler implicit move (Arthur O'Dwyer) (C++23?)
  - P1663: Supporting return-value-optimisation in coroutines (Lewis Baker)
  - P1906: Provided operator= return lvalue-ref on rvalue (Peter Sommerlad)
  - P2307: Lvalue closures (Jens Gustedt)
  - P1946: Allow defaulting comparisons by value (Barry Revzin, Casey Carter)
  - P2027: Moved-from objects need not be valid (Geoff Romer)
  - P2226: A function template to move from an object and reset it (…) (Giusepper D'Angelo)
  - P2012: Fix the range-based for loop (Various Authors)

References (☺)

- Books, Talks & Blogs:

  - Book: David Vandevoorde, Nicolai Josuttis, Douglas Gregor: C++ Templates: The Complete Guide
  - Paper: Bjarne Stroustrup: “New” Value Terminology
  - Talk: Ben Deane: Deducing this Patterns (CppCon 2021)
  - Talk: Peter Sommerlad: Reducing immediate dangling? (lightning talk) (CppCon 2019)
  - Talk(s): Walter E.Brown: Modern Template Metaprogramming: A Compendium (CppCon 2014)
  - Blog: Barry Revzin: xvalues and prvalues: The Next Generation (Discussing Identity function safely)
  - Blog: Barry Revzin: Value Categories in C++17
  - Blog: Tristan Brindle: Rvalue Ranges and Views in C++20
  - Blog: Thomas Becker: C++ Rvalue References Explained
  - StackOverflow: Do rvalue references to const have any use
  - Manual: Rvalue references in Chromium

"The best number is 73. Why? 73 is the 21st prime number. Its mirror, 37, is the 12th and its mirror, 21, is the product of multiplying 7 and 3."
"We get it, 73 is the Chuck Norris of numbers!"
"Chuck Norris wishes. In binary 73 is a palindrome, 1001001, which backwards is 1001001. All Chuck Norris backwards gets you is Sirron Kcuhc!"'